

A dual porosity model of nutrient uptake by root hairs

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Summary

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• The importance of root hairs in the uptake of sparingly soluble nutrients is understood qualitatively, but not quantitatively, and this limits efforts to breed plants tolerant of nutrient-deficient soils.

• Here, we develop a mathematical model of nutrient uptake by root hairs allowing for hair geometry and the details of nutrient transport through soil, including diffusion within and between soil particles. We give illustrative results for phosphate uptake.

• Compared with conventional 'single porosity' models, this 'dual porosity' model predicts greater root uptake because more nutrient is available by slow release from within soil particles. Also the effect of soil moisture is less important with the dual porosity model because the effective volume available for diffusion in the soil is larger, and the predicted effects of hair length and density are different.

• Consistent with experimental observations, with the dual porosity model, increases in hair length give greater increases in uptake than increases in hair density per unit main root length. The effect of hair density is less in dry soil because the minimum concentration in solution for net influx is reached more rapidly. The effect of hair length is much less sensitive to soil moisture.

Introduction

Plant uptake of nutrient ions that are strongly sorbed on soil particles – such as phosphate – tends to be limited by diffusion through the soil to root surfaces. Hence uptake is more often limited by root architecture and the foraging potential of the root system than by the activities of root ion transporters (Tinker & Nye, 2000). Among root architecture traits, root hairs have been shown to be particularly important. This has been demonstrated by studies with autoradiography of soil depletion zones (Bhat & Nye, 1973), with mutants (e.g. Bates & Lynch, 2000; Gahoonia & Nielsen, 2003) and with recombinant inbred lines (e.g. Yan *et al.*, 2004; Wang *et al.*, 2004; Pariasca-Tanaka *et al.*, 2009; Zhu *et al.*, 2010). However, we do not have a good quantitative understanding of the effects of hair geometries and soil characteristics under different environmental conditions on nutrient uptake by root hairs. This

means we do not know, for example, the pay-off between hair density and length, and how this differs between environments.

It is well known that root hair development varies greatly with internal (e.g. tissue phosphate concentration) and external (e.g. pH and Al^{3+} concentration) conditions and between species (Tinker & Nye, 2000). Dittmer (1949) and Caradus (1980) found in a study of 50 plant species that root hair length ranged from 0.08 to 1.5 mm, and Itoh & Barber (1983) and Mackay & Barber (1985) found differences in both the length (0.04–0.6 mm) and density (200–1819 cm^{-1}) of root hairs among six contrasting plant species. Phosphate and soil moisture are key regulators of both root hair length and density (Tinker & Nye, 2000). However, in general it is difficult to control hair length and density independently, although there have been efforts to do this with mutants and recombinant inbred lines (Bates & Lynch, 2000; Gahoonia & Nielsen, 2003). So we lack experimental evidence for the relative importance of these for uptake in different environments.

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To address this issue theoretically, Leitner *et al.* (2010a) developed a model of nutrient uptake by root hairs in soil, with a simple treatment of the transport processes controlling delivery of nutrient ions through soil to absorbing root surfaces. Leitner *et al.* (2010a) made the simplifying assumption that the soil around an individual root hair is effectively homogeneous at the scale of the hair. However, this assumption is unlikely to be valid given that typical inter-hair distances (a few tens of μm) are comparable to the diameters of individual soil micro-aggregates. This will have a bearing on the effect of hair geometry on uptake and its interactions with soil conditions.

In this paper, we develop a model that allows explicitly for the soil micro-structure and root hair geometry, with which to explore such effects. We allow for transport in the soil solution within and between soil particles, with simultaneous reactions on the external and internal surfaces of soil particles. In Leitner *et al.*'s (2010a) model, nutrient transport through the soil was described with a single porosity diffusion model. Equilibration between the soil solution and soil surfaces on minerals and organic matter was assumed to be instantaneous compared with transport through the soil bulk, and was described with a constant buffer power (i.e. constant solid-to-solution partition coefficient). This is adequate for weakly sorbed solutes. However, for more strongly sorbed solutes, such as phosphate and

many micro-nutrients, equilibration between the soil solid and solution can be slow, often being limited by slow diffusion from sorption sites within soil particles (Ptashnyk *et al.*, 2010). In such cases single porosity models can lead to seriously misleading results (Ptashnyk *et al.*, 2010) and 'dual porosity' models are more appropriate. In this paper, we combine the approaches of Leitner *et al.* (2010a) and Ptashnyk *et al.* (2010) to allow for dual porosity behaviour in root hair zones. With the model that we develop we are able to quantify the influence of environmental conditions on nutrient uptake by hairy roots, allowing for effects at the scale of individual root hairs and soil pores. Compared with other models of nutrient uptake by root hairs (e.g. Bhat *et al.*, 1976; Itoh & Barber, 1983; Ma *et al.*, 2001), the advantage of our model is that the sink term describing the uptake by hairs is derived rigorously from micro-scale considerations, allowing for soil heterogeneity at the scale of the individual hair and soil particle to be accounted for.

We use the model to explore a wide range of values of the main root and soil variables involved, so as to identify optimal root geometries for uptake of strongly sorbed nutrients. Specifically we explore: the relative effects of root hair density and length on uptake; the effects of soil water content on the importance of root hairs; and the effects of slow release of sorbed nutrient from within soil particles on overall uptake. The notation used is given in Table 1.

Table 1 Nomenclature

Symbol	Description
x	Macroscopic space variable
y	Microscopic space variable
r	Radial distance inside soil particle (cm)
l	Length of unit cell containing a soil particle, associated extra-particle solution and gas, and root hairs (cm)
L	Length of root with root hairs (cm)
L_h	Length of root hairs (cm)
L_x	Length of zone of root influence (cm)
a	Radius of soil particle (cm)
a_r	Radius of the main root (cm)
a_h	Radius of root hair (cm)
C_e and C_i	Concentration of nutrient in extra- and intra-particle solutions ($\mu\text{mol cm}^{-3}$)
S_e and S_i	Concentration of nutrient on extra- and intra-particle surfaces per unit solid mass ($\mu\text{mol g}^{-1}$)
k_{Fe} and k_{Be}	Rate constants for backward and forward sorption reactions on extra-particle surfaces (s^{-1})
k_{Fi} and k_{Bi}	Rate constants for backward and forward sorption reactions on intra-particle surfaces (s^{-1})
β	Coefficient in equilibrium sorption equation ($\text{cm}^3 \text{g}^{-1}$)
θ	Water content of whole soil per unit soil volume ($\text{cm}^3 \text{cm}^{-3}$)
θ_e and θ_i	Water content of extra- and intra-particle space per unit space volume (with hat, per unit whole soil volume) ($\text{cm}^3 \text{cm}^{-3}$)
ρ_s , ρ_i and ρ_s	Soil solid mass per unit whole soil, particle and solid volume (g cm^{-3})
ϕ_e , ϕ_i	Porosity of extra- and intra-particle space per unit whole soil volume ($\text{cm}^3 \text{cm}^{-3}$)
σ_e and σ_i	Soil mass per unit external and internal surface area of particle (g cm^{-2})
D_i	Nutrient diffusion coefficient in intra-particle pores ($\text{cm}^2 \text{s}^{-1}$)
D_l	Nutrient diffusion coefficient in free solution ($\text{cm}^2 \text{s}^{-1}$)
f_e and f_i	Nutrient diffusion impedance factor for extra- and intra-particle pores
E_h and E_r	Nutrient efflux parameter for root hair and root ($\mu\text{mol cm}^{-2} \text{s}^{-1}$)
F_h and F_r	Maximal nutrient influx into root hair and root ($\mu\text{mol cm}^{-2} \text{s}^{-1}$)
K_h and K_r	Michaelis–Menten constant for root hair and root nutrient uptake ($\mu\text{mol cm}^{-3}$)

Description

Dimensional model explicitly considering root hair geometry

Following Leitner *et al.* (2010a), we consider a single hairy root in soil. The primary region of interest is the root hair zone, extending 2–4 cm back from the root tip in most crop plants, where growth of the main root axis has stopped and root hairs have emerged to different extents. We treat the soil as a doubly porous medium, with an extra-particle pore space containing solution and air, and an intra-particle space also containing solution and air. We make the following approximations.

- We consider diffusion to be the main transport process, that is, we ignore convection, as is realistic for strongly sorbed solutes on the single root scale (Roose *et al.*, 2001; Roose & Kirk, 2009).
- We consider Cartesian geometry rather than the more computationally complex cylindrical geometry. This is justified as long as the distance between hairs at the root surface is comparable to the distance between hair tips, and as long as the timescale of nutrient uptake is faster than the diffusion timescale. This was shown to be the case for most strongly sorbed solutes by Roose (2000).
- We consider a periodic domain containing cylindrical root hairs orthogonal to a planar root surface, with at least one soil particle between adjacent root hairs (Supporting Information Fig. S2). Outside the root hair zone, the periodic domain contains only soil particles. The distance between hairs (which is comparable to the diameter of the soil particle) defines the microscopic length scale, l , the root length defines the macroscopic length, L . The model requires that there is a separation of length scales, that is, $l \ll L$.

Note that although, as a special case, we consider only a single soil particle between adjacent root hairs, the model could also be solved for multiple particles between hairs. However, with multiple particles, the soil approximates a quasi-homogeneous body (Tinker & Nye, 2000), as simulated by Leitner *et al.* (2010a). In our model the soil particles represent discrete zones of porous soil material, which are smaller than the 0.1–1-mm space between adjacent root hairs. In conventional, single porosity models, soil particles at this scale are treated as a solid phase, on and in which the nutrient is sorbed. However, in reality they are aggregations of individual soil constituents joined together by organic matter and iron (Fe) and aluminium (Al) oxyhydroxide coatings, or they are single mineral phases containing micro-pores and cracks.

In the treatment below, we use homogenization techniques to derive a dual porosity model as described by Ptashnyk *et al.* (2010) and Ptashnyk & Roose (2010). The dimensional equations on which the homogenized model is based are as follows. We distinguish between the solute concentration in solution between the soil particles, C_e , and

that inside the particle, C_i , and we take into account the concentrations on the external and internal particle surfaces, S_e and S_i , respectively. The continuity equation for the extra-particle space is:

$$\frac{\partial C_e}{\partial t} = \nabla \cdot (D_l \nabla C_e) \quad \text{Eqn 1}$$

where D_l is the solute diffusion coefficient in free solution, and that for the intra-particle space:

$$\frac{\partial(\theta_i C_i)}{\partial t} = \nabla \cdot (D_i \nabla C_i) - \frac{\partial(\rho_i S_{i1})}{\partial t} - \frac{\partial(\rho_i S_{i2})}{\partial t} \quad \text{Eqn 2}$$

where the last two terms allow for fast and slow reactions inside the particle, and the diffusion coefficient $D_i = \theta_i f_i D_l$, where f_i is an impedance factor which takes account of the tortuosity of the diffusion pathway inside the particle resulting from geometric and electrostatic effects. In effect, we have already homogenized the transport and sorption reaction processes inside the particle and the effect of the complicated geometry is expressed in D_i , which represents the impeded diffusion within the particle without actually taking into account the micro-pore geometry.

The two equations for the extra- and intra-particle concentrations need to be coupled via boundary conditions at the particle surface. The boundary conditions showing the solute flux and concentration continuity at the particle surface are

$$D_l \nabla C_e \cdot v = D_i \nabla C_i \cdot v - \frac{\partial(\sigma_e S_{e1})}{\partial t} - \frac{\partial(\sigma_e S_{e2})}{\partial t} \quad \text{Eqn 3}$$

where v is a unit vector normal to the surface pointing outwards, and

$$C_e = C_i \quad \text{Eqn 4}$$

The last two terms in Eqn 3 allow for fast and slow sorption reactions on the external particle surface. We represent the sorption–desorption reactions on internal and external particle surfaces with simple linear kinetics

$$\frac{\partial(\rho_i S_i)}{\partial t} = k_{Fi} \hat{\theta}_i C_i - k_{Bi} \rho_i S_i \quad \text{Eqn 5}$$

and

$$\frac{\partial(\sigma_e S_e/l)}{\partial t} = k_{Fe} \hat{\theta}_e C_e - k_{Be} \sigma_e S_e/l \quad \text{Eqn 6}$$

where k_F and k_B are forward and backward rate constants (omitting subscripts 1 and 2 for fast and slow reactions). We use linear kinetics because they give the simplest

realistic representation of sorption–desorption. Any nonlinear kinetics can be approximated by the linear kinetics at the leading order.

We choose the following boundary conditions.

- (1) Extra particle gas spaces: we apply a zero flux boundary condition

$$D_l \nabla C_e \cdot \mathbf{v} = 0 \quad \text{Eqn 7}$$

- (2) Root hair surface: uptake is described by a function g_h such that

$$-D_l \nabla C_e \cdot \mathbf{v} = g_h \quad \text{Eqn 8}$$

- (3) Root surface: uptake is described by a function g_r such that

$$-D_l \nabla C_e \cdot \mathbf{v} = g_r \quad \text{Eqn 9}$$

where $\mathbf{v}$ is the outward vector normal to the surface.

We take the root uptake function g_r to be of Michaelis–Menten type (Tinker & Nye, 2000, and references therein). As there is no equivalent information for uptake by root hairs, we assume that a similar relationship applies. Hence,

$$g_h = \frac{F_h C_e}{K_h + C_e} - E_h \quad \text{and} \quad g_r = \frac{F_r C_e}{K_r + C_e} - E_r \quad \text{Eqn 10}$$

where F , K and E are the maximal nutrient influx, the Michaelis–Menten constant and efflux, respectively; and subscripts ‘h’ and ‘r’ denote root hair and root values, respectively. We could have used different forms of g_r and g_h without any significant influence on the results as long as the order of magnitude for the uptake functions does not change and as long as the functions remain sublinear, that is, they do not blow up as the surface concentration changes (Ptashnyk & Roose, 2010).

Homogenized model

In this section, we present an effective macroscopic model derived from the above equations (see the Supporting Information Notes S1, S2 for full details). In outline, we derived the model using homogenization theory, which allows for the cumulative effect of root hairs without explicitly considering individual hairs, while still taking into account the detail of root hair and soil particle geometry and properties. The full homogenized model derivation can be found in the Notes S2 (Eqns S6–S9). We present a simplified version of the homogenized model with two assumptions: first, that the fast reactions are effectively instantaneous compared with diffusion through the soil solution; secondly, that the soil particles are spherical, resulting in only radial diffusion gradients within the particle, that

is, one-dimensional spherically symmetric diffusion within the particle. Similarly, in the macro-scale we only take into account the radial distance from the main root (considering Cartesian geometry following Roose, 2000). Such a simplification in the macro-scale is valid because there will only be large diffusion gradients in the radial direction. Hence Leitner *et al.* (2010b) found that the predictions for nutrient uptake of one-dimensional effective models (as here) agreed with those of fully three-dimensional models.

We consider two zones: the hair zone, where there is a sink term for the effect of root hairs in the equation for the extra-particle space:

$$\begin{aligned} (\theta_e^h + b_e^h) \partial_t C_e &= \partial_x (D_l \theta_e^h f_e^h \partial_x C_e) \\ &- \frac{4\pi a^2}{l^2} \left\{ \frac{D_i}{l} \partial_r C_i \Big|_{r=a} + \left(k_{Fe} \hat{\theta}_i C_i - \frac{k_{Be} \sigma_e}{l} S_e \right) \right\} \\ &- \frac{2\pi a b_h}{l} \left\{ \frac{1}{l} \left(\frac{F_h C_e}{K_h + C_e} - E_h \right) \right\} \end{aligned} \quad \text{Eqn 11}$$

and the zone beyond the root hair zone, where there is no sink term:

$$\begin{aligned} (\theta_e + b_e) \partial_t C_e &= \partial_x (D_l \theta_e f_e \partial_x C_e) \\ &- \frac{4\pi a^2}{l^2} \left\{ \frac{D_i}{l} \partial_r C_i \Big|_{r=a} + \left(k_{Fe} \hat{\theta}_i C_i - \frac{k_{Be} \sigma_e}{l} S_e \right) \right\} \end{aligned} \quad \text{Eqn 12}$$

In Eqn 11, θ_e^h , f_e^h and b_e^h are superscripted ‘h’ to indicate the root hair zone. The buffer power, b_e^h , is given by

$$b_e^h = \frac{4\pi a^2 \theta_e^h k_{Be}}{l^2 k_{Fe}} \quad \text{Eqn 13}$$

The boundary conditions for C_e are:

$$D_l f_e^h \partial_x C_e = \frac{F_r C_e}{K_r + C_e} - E_r, \quad \text{at } x = 0 \quad \text{Eqn 14}$$

and

$$\partial_x C_e = 0 \quad \text{at } x = L_x \quad \text{Eqn 15}$$

where L_x is the length of the zone of root influence (bounded by the mid-point between neighbouring roots). We also have continuity of concentrations and fluxes on the root hair zone boundary. Notice that both the effective equations for the extra-particle concentration are coupled to the micro-scale concentration C_i , which at every x position satisfies the equation

$$(\theta_i + b_i) \partial_t C_i = D_i / r^2 \partial_r (r^2 \partial_r C_i) - (k_{Fi} C_i - k_{Bi} \rho_i S_i) \quad \text{Eqn 16}$$

where b_i is the buffer power, given by

$$b_i = \frac{\hat{\theta}_i k_{Bi}}{k_{Fi}} \quad \text{Eqn 17}$$

and C_i is coupled to C_e via the boundary condition $C_i = C_e$ at $r = a$.

The model described here contains terms for both the cumulative effect of root hairs and the flux from within the particles; we refer to this as the dual porosity model. Depending on the relative strengths of these terms, we could further simplify the model as in Leitner *et al.* (2010a). If root hair uptake is weak, the term $2\pi a_b/l^2 \{F_b C_e/(K_b + C_e) - E_b\}$ in Eqn 11 can be ignored; if root hair uptake is very strong, Eqn 11 can be dispensed with altogether and uptake calculated for an effective root radius at the outer boundary of the root hair zone using a zero sink approach at that boundary. Similarly, if the diffusion impedance within the soil particle is very strong, we could ignore the coupling between the micro- and macro-scales in our equations just as in Ptashnyk *et al.* (2010).

Sensitivity analysis

Having developed the model, we can now investigate the influence of root and soil variables on nutrient uptake by hairy roots. As a starting point, we compare uptake by roots with and without root hairs, and we consider our dual porosity model against a single porosity model of the type used by Leitner *et al.* (2010a). In this case, the single porosity model is given by

$$(\theta_e^b + b_e^b) \partial_t C_e = \partial_x (D_l f_e^b \partial_x C_e) - 2\pi a_b l^2 \left(\frac{F_b C_e}{K_b + C_e} - E_b \right) \quad \text{Eqn 18}$$

in the root hair zone, and

$$(\theta_e + b_e) \partial_t C_e = \partial_x (D_l f_e \partial_x C_e) \quad \text{Eqn 19}$$

outside the root hair zone.

A dilemma in comparing the single and dual porosity models is how to distribute the additional nutrient associated with the intra-particle space. For the same total nutrient concentration and the same concentration in solution in the extra-particle pores, the sorbed concentration will differ between the models if b_e is the same. Alternatively, b_e could be increased in the single porosity model so that the sorbed concentration is the same. In our sensitivity analysis, we compare the full dual porosity model (for which $b_e = 9.1$) with a single porosity model in which the initial concentrations on external soil surfaces and in solution are the same as in the dual porosity model (hence $b_e = 9.1$). An alternative single porosity model is one in which the total concentration in the soil (external and inter-

nal surfaces) is the same (here $b_e = 11\,168$). We refer to these models as Single 1 and Single 2, respectively.

Another important variable in nutrient uptake by root hairs is soil moisture (Mackay & Barber, 1985; Sangakkara *et al.*, 1996). The effect of root hairs is likely to be particularly important in dry soils because the impedance to diffusion is significantly greater. We examine the effect of moisture content by changing the extra- and the intra-particle water contents: $\hat{\theta}_e$ and $\hat{\theta}_i$. This implies changes in diffusion impedance factors in the extra- and intra-particle spaces: f_e and f_i . Note that changing $\hat{\theta}_e$ entails re-solving the appropriate cell problem to find the new effective diffusion coefficient for any given soil particle and root hair geometry (Notes S3), whereas changes in $\hat{\theta}_i$ can be allowed for with the formula $D_i = \hat{\theta}_i f_i D_l$ because we do not have reliable information about the soil particle micro-structure. If the air volume is the same in the root hair zone and away from it, θ_x will be $< \theta_b$ to account for the root hair volume. However, we make θ_x and θ_b equal, implying that the root hairs occupy some of the air space. Thus $\theta_a = \theta_b = \hat{\theta}_e$.

We also consider the effects of root hair geometry as represented by the length of the hair zone, L_h , and the number of hairs per unit length of main root, N . Following Leitner *et al.* (2010a), we define the inter-hair distance, l , as equal to the square root of the main root surface area associated with a single root hair, that is, $l = \sqrt{2\pi a_r/N}$ or $N = 2\pi a_r/l^2$ where a_r is the root radius. Using this we reformulate Eqn 10 as

$$\begin{aligned} (\theta_e^b + b_e^b) \partial_t C_e &= \partial_x (D_l \theta_e^b f_e^b \partial_x C_e) \\ &- \frac{4\pi a^2}{l^2} \left\{ \frac{D_i}{l} \partial_r C_i \Big|_{r=a} + \left(k_{Fe} \hat{\theta}_i C_i - \frac{k_{Be} \sigma_e}{l} S_e \right) \right\} \\ &- N \frac{a_b}{a_r} \left\{ \left(\frac{F_b C_e}{K_b + C_e} - E_b \right) \right\} \end{aligned} \quad \text{Eqn 20}$$

where a_h is the root hair radius. In reformulating Eqn 11 in this way we assume that the inter-hair distance is equal to the length of the unit cell containing a soil particle. If we now relax this assumption, Eqn 20 is still valid but with a different impedance factor f_e^b , as the cell problem to be solved is different. However, because the root hairs occupy a small volume fraction of the cell, they do not affect the impedance factor f_e^b beyond the third decimal place and thus we ignore this. The presence of root hairs in the intra-particle space only changes the effective diffusion impedance by $c. 0.1\%$ (Notes S3), so the effect of root hairs on diffusion impedance can be ignored.

Parameter values

The standard parameter values for the simulations are given in Table 2. They were derived as follows. The soil

parameters are the same as in Ptashnyk *et al.* (2010). The derivation of the other parameters is discussed below.

Geometry The relationships among the soil bulk density, water content and gas content are constrained by the model geometry. The total soil porosity is given as

$$\phi = \hat{\phi}_e + \hat{\phi}_i = 1 - \rho/\rho_s$$

where $\hat{\phi}_e$ and $\hat{\phi}_i$ are the porosities of extra- and intra-particle spaces, respectively, and, ρ_s is the density of the soil solid. The upper limit on $\hat{\phi}_e$ as the thickness of the liquid layer around a particle tends to zero is $(l^3 - 4/3\pi a^3)/l^3 = 0.476$. In order to ensure the existence of a thin liquid layer around a particle, we choose $\hat{\phi}_e = 0.48$, that is, the radius of soil particles, $a = 9.98 \times 10^{-3}$ cm.

If the air spaces are spherical, there is a lower limit on the soil water content of $\theta_e = 0.14$. Therefore, to model drier soils we must allow for nonspherical air spaces. This does not change the validity of the model. The geometry of the air space is represented in the model through the value of the extra-particle impedance factor, f_e (see next section, Diffusion). We have calculated values of f_e for different shapes (spherical and elliptical) and positions of air spaces within the unit cell containing a soil particle (see the Supporting Information Fig. S1), using standard routines for this in the mathematical software COMSOL MULTIPHYSICS (<http://www.comsol.com/>). We found that the total volume of the air space was far more important than the number, shape or position of the separate air spaces that it was divided into. This should also be true for irregularly shaped, individual, connected spaces. We conclude that the geometry of the air space within the unit cell is not crucial to the calculations.

Table 2 Parameter values used for simulations; all other parameters are derived from these as described in the text

Parameter	Value
a	9.98×10^{-3} cm
l	0.02 cm
L	1 cm
L_h	0.08 cm
L_x	1 cm
a_h	5×10^{-4} cm
a_r	10^{-2} cm
ρ	1.1 g cm $^{-3}$
σ_i	5×10^{-4} g cm $^{-2}$
β_1	500 cm 3 g $^{-1}$
β_2	1500 cm 3 g $^{-1}$
k_{F2}	10^{-4} s $^{-1}$
D_i	9×10^{-6} cm 2 s $^{-1}$
E_h and E_r	9.6×10^{-9} μ mol cm $^{-2}$ s $^{-1}$
F_h and F_r	2.2×10^{-7} μ mol cm $^{-2}$ s $^{-1}$
K_h and K_r	2.3×10^{-3} μ mol cm $^{-3}$

We consider three moisture cases with the values of $\hat{\theta}_i$, and $\hat{\theta}_e$ given in Table 3. These cover the range from about field capacity to permanent wilting point for a sandy loam. We assume that the intra-particle space is always water-saturated, that is, $\hat{\theta}_i = \hat{\phi}_i$, and as the soil is dried from field capacity, water is removed solely from the extra-particle space, as is realistic. The air volume fraction in the extra-particle space varies between 0.114 ($\hat{\theta}_e = 0.366$) and 0.386 ($\hat{\theta}_e = 0.094$), thus the soil is never fully saturated.

The specific surface area of the external particle surface is $SSA_e = 4\pi a^2/(l^3\rho)$, that is, $\sigma_i = 1/SSA_e = 7.03 \times 10^{-3}$ g cm $^{-2}$. In general, the specific surface area of the internal particle surface (SSA_i) depends on how the total internal porosity is distributed among pores of different sizes and is a function of the composition of the particle as well as its total porosity. Typical values are $SSA_i = 500$ – $15\,000$ cm 2 g $^{-1}$, that is, $\sigma_i = 2 \times 10^{-3}$ to 6.7×10^{-5} g cm $^{-2}$.

The size and density of hairs on the root vary greatly among species and with environmental factors, including soil moisture and nutrient status (Tinker & Nye, 2000). Based on the references given in the Introduction, we take $L_h = 0.08$ mm as standard and consider the range 0.02–0.16 cm. In data reviewed by Leitner *et al.* (2010a) hair density N varied from 10 to 1200 cm $^{-1}$ of root length in onion (*Allium*), from 200 to 600 cm $^{-1}$ in various members of the Gramineae, and up to 1700 cm $^{-1}$ in certain dicots. Based on this, we take $N = 158$ cm $^{-1}$ as standard and consider the range from half to three times this. Also from the data in Leitner *et al.* (2010a) we take $a_h = 5$ μ m. We take for the length of root with hairs $L = 1$ cm and the total length of the zone of root influence $L_x = 1$ cm.

Diffusion We now consider the effect of the geometry on the impedance to diffusion inside and outside the soil particle. The diffusion coefficient in the intra-particle space is $D_i = \theta_i f_i D_1$, where θ_i is the water content in the intra-particle space per unit particle volume: $\theta_i = \hat{\theta}_i/(1 - \hat{\phi}_e)$. We use $D_1 = 9 \times 10^{-6}$ cm 2 s $^{-1}$ (the value for H $_2$ PO $_4^-$) and take the intra-particle impedance factor to be $f_i = 0.001$ as estimated by Ptashnyk *et al.* (2010) after Nye & Staunton (1994). To calculate the impedance factor for the extra-particle

Table 3 Water contents of extra- and intra-particle spaces for the three cases simulated and corresponding diffusion impedance factors

	Case 1	Case 2	Case 3
$\hat{\theta}_e$	0.366	0.274	0.094
$\hat{\theta}_i$	0.105	0.105	0.105
$\hat{\phi}_e - \hat{\theta}_e$	0.114	0.206	0.386
f_e^h (calculated)	0.680	0.621	0.532
f_e (calculated)	0.689	0.628	0.542
f_e	0.366	0.274	0.094
f_i	0.001	0.001	0.001

The calculated impedance factors only allow for geometric effects (see text).

space, we solve the associated cell problem in and away from the root hair zone. Table 3 gives the results of these calculations. Note that for different configurations of the gas spaces (different shapes or positions) we would have slightly different values for the extra-particle impedance factor, but they will always be of the same order of magnitude. The values of f_e so calculated are larger than experimentally determined values (Tinker & Nye, 2000), particularly at smaller moisture contents. This is because the calculations only allow for geometric effects, but not for the electrostatic interactions between soil surfaces and diffusing ions, which result in ion exclusion from narrow pores by negative adsorption, and increased viscosity of water near charged surfaces (Tinker & Nye, 2000). These effects are increasingly important at smaller moisture contents. Based on experimentally determined impedance factors, the relation $f_e = \theta'_e$ is realistic (Olesen *et al.*, 2001), and we use that for the main calculations below.

Sorption We now relate the parameters for sorption on internal and external surfaces to the corresponding, measurable parameters for the whole soil (cf. Ptashnyk *et al.*, 2010). Dropping subscripts 1 and 2 for fast and slow reactions, we have:

$$\frac{\partial(\rho S)}{\partial t} = k_F \theta C - k_B \rho S$$

At equilibrium $\partial S / \partial t = 0$ and therefore $S = \beta C$ where $\beta = k_F \theta / (k_B \rho)$.

We assume that the density of sorption sites on internal and external surfaces is the same ($\sigma_i S_i = \sigma_e S_e$). Also $k_{Bi} = k_{Be} = k_B$. Using the fact that $S = S_e + S_i$ we obtain for the reactions on internal surfaces:

$$k_{Fi} = k_F \frac{\theta \rho_i}{\theta_i \rho (1 + \sigma_i / \sigma_e)}$$

and

$$\beta_i = \frac{k_{Fi} \hat{\theta}_i}{k_{Bi} \rho_i} = \frac{\beta}{1 + \sigma_i / \sigma_e}$$

Similarly for the reactions on external surfaces:

$$k_{Fe} = k_F \frac{\theta \sigma_e}{\theta'_e l \rho (1 + \sigma_e / \sigma_i)}$$

and

$$\beta_e = \frac{k_{Fe} \hat{\theta}_e l}{k_{Be} \sigma_e} = \frac{\beta}{1 + \sigma_e / \sigma_i}$$

If we now denote the ratio of rapid to slow reaction sites $\gamma = S_1 / S_2 = \beta_1 / \beta_2$ we obtain the following forward reaction rates:

$$k_{F1}^e = k_{F1} \frac{\theta \sigma_e}{\hat{\theta}_e l \rho (1 + 1/\gamma)(1 + \sigma_e / \sigma_i)}$$

$$k_{F1}^i = k_{F1} \frac{\theta \rho_i}{\hat{\theta}_i \rho (1 + 1/\gamma)(1 + \sigma_i / \sigma_e)}$$

$$k_{F2}^e = k_{F2} \frac{\theta \sigma_e}{\hat{\theta}_e l \rho (1 + 1/\gamma)(1 + \sigma_e / \sigma_i)}$$

$$k_{F2}^i = k_{F2} \frac{\theta \rho_i}{\hat{\theta}_i \rho (1 + 1/\gamma)(1 + \sigma_i / \sigma_e)}$$

while the different buffer powers are given by:

$$\beta_{e1} = \frac{\beta_1}{(1 + 1/\gamma)(1 + \sigma_e / \sigma_i)} \quad \beta_{i1} = \frac{\beta_1}{(1 + 1/\gamma)(1 + \sigma_i / \sigma_e)}$$

$$\beta_{e2} = \frac{\beta_2}{(1 + 1/\gamma)(1 + \sigma_e / \sigma_i)} \quad \beta_{i2} = \frac{\beta_2}{(1 + 1/\gamma)(1 + \sigma_i / \sigma_e)}$$

Note we have assumed $k_{B1}^e = k_{B1}^i = k_{B1}$ and $k_{F2}^e = k_{F2}^i = k_{F2}$, because we assume the processes inside the particles are similar to the processes on external surfaces. This is reasonable because we expect the average composition of the soil particles not to have any directional effects. With this notation we have $b_e = 4\pi\sigma_e a^2 \beta_{e1} / l^3$ and $b_i = \rho_i \beta_{i1}$. We take as standard the following values for sorption of phosphate on a sandy loam soil (cf. Ptashnyk *et al.*, 2010; after Nye & Staunton, 1994): $\beta_1 = 500 \text{ cm}^3 \text{ g}^{-1}$, $\beta_2 = 1500 \text{ cm}^3 \text{ g}^{-1}$ and $k_{F2} = 10^{-4} \text{ s}^{-1}$.

Root and root hair uptake properties We know of no published data on the phosphate uptake parameters of individual root hairs, and we therefore use the same values as for the main roots. We consider this is realistic because the uptake properties of individual root cells will depend on the properties of the same membrane transporters (Mudge *et al.*, 2002). There is no reason to believe that these will differ between the main root and root hairs. However, the densities of transporters may vary, potentially resulting in differences in net uptake properties (Gilroy & Jones, 2000). As we do not have a good quantitative understanding of this, we will not consider it further. However, it would be straightforward to include such a phosphate-transporter distribution effect in the model by varying the root hair uptake parameters. The values of the uptake parameters in Table 2 are from Föhse *et al.* (1991) and are typical of phosphate uptake by moderately phosphate-starved grass species.

Results

Root phosphate acquisition in the single and dual porosity models

We illustrate the differences between the models using the standard parameter values in Table 2 and the three water

Table 4 Effect of root hairs and soil moisture content (Cases 1–3 in Table 3) on total phosphate uptake calculated with the dual porosity and two single porosity models

	Cumulative uptake (nmol cm ⁻² main root in 14 d)		Relative effect (%)
	With root hairs	No root hairs	
<u>Case 1</u> ($\theta = 0.471$)			
Dual	34.4 (40.4)	14.2 (21.5)	141 (87)
Single 1	10.5 (13.1)	7.4 (10.3)	41 (27)
Single 2	101.8 (114.4)	17.7 (29.5)	475 (288)
<u>Case 2</u> ($\theta = 0.379$)			
Dual	29.5 (35.9)	8.7 (16.0)	236 (123)
Single 1	8.2 (11.1)	5.0 (8.1)	64 (37)
Single 2	94.3 (105.5)	9.9 (20.3)	846 (418)
<u>Case 3</u> ($\theta = 0.199$)			
Dual	20.1 (26.5)	1.3 (5.7)	1500 (358)
Single 1	3.9 (6.8)	0.9 (3.5)	321 (94)
Single 2	85.6 (91.5)	1.2 (6.0)	7133 (1410)

Values in parentheses are calculated with the impedance factors only allowing for geometric effects (Table 3). The relative effect is the percentage change in uptake with root hairs. Time = 14 d; amount of labile phosphate in soil in zone of root influence initially = 25.87 μmol ; other parameter values as in Table 2.

contents in Table 3. Table 4 gives the cumulative uptake calculated with the three models at different moisture contents with and without root hairs. Note that in Single 1 the concentration on external surfaces is the same as in the dual porosity model; in Single 2 the total concentration in the soil is the same, that is, the plant has access to phosphate inside the soil particles.

It will be seen that, in all the three cases, root hairs greatly increase phosphate uptake, as expected. The effect is more pronounced with Single 2 (same total concentration) because more phosphate is available to the plant almost instantaneously as there is no internal particle diffusion pathway. Comparing the dual porosity model with Single 1 (same concentration on external surfaces), the overall effect of root hairs is greater with the dual porosity model because the root system is able to take up phosphate that is slowly released from within soil particles. Comparing the two single porosity models, cumulative phosphate uptake is far greater with Single 2 than Single 1, with and without root hairs. This is expected given that the effective buffer power in Single 2 is much greater than that in Single 1, resulting in greater availability of phosphate through desorption from external particle surfaces into the soil solution and thus steeper phosphate depletion zones (see Fig. 1, discussed in the next paragraph). The results in Table 4 show that the relative importance of root hairs for phosphate uptake increases as the moisture content decreases. The overall effect of moisture is greatest with Single 1 and smallest with Single 2. Table 4 also shows the effect of the diffusion impedance factor. The impedance factors only allowing for geometric

effects are larger, and as a result uptake is less limited by diffusion. Hence, in Table 4, uptake is larger with these impedance factors, and the relative effects of root hairs are smaller, especially at low moisture content where diffusion is increasingly limiting.

Fig. 1 shows the corresponding concentration profiles with distance from the main root axis. The left-hand panels show the concentrations of phosphate in the extra-particle soil solution; the right-hand panels show the corresponding changes in the whole-soil concentration, that is, the net depletion from the soil as a result of uptake by the root. For all three models, as expected, the effect of root hairs is to lower the concentration in solution in the root hair zone. The decrease is smaller for Single 2 because the concentration in solution is better buffered by desorption from the soil solid. Comparing the dual porosity model with Single 1, the solution concentration with the dual porosity model is greater because of buffering by slow release of phosphate from within soil particles. The profiles of change in total concentration in the soil (right-hand panels in Fig. 1) differ between the models both in the extent of depletion close to the root surface and in the spread of the depletion zone away. The net depletion increases and the spread decreases in the order Single 1, dual, Single 2, consistent with the order of increasing buffer power.

Effect of root hair geometry on phosphate acquisition

We now use the dual porosity model to assess the effects of root hair geometry and the relative benefits to a plant with longer root hairs vs greater numbers of root hairs. We consider the effects of the length of the root hair zone (L_h) and the number of root hairs per unit length of the main root (N) using Eqn 19 as our model and $N = 2\pi a_r / l^2$ (see paragraph preceding Eqn 19). Fig. 2 gives the time courses of uptake calculated with the dual porosity model over 14 d for a range of L_h at constant N , and vice versa. It shows that, in all cases, cumulative uptake increases sharply over the first day and then continues to increase at a gradually declining rate over subsequent days as phosphate is depleted around the root. For the combinations of L_h and N values considered, a 4-fold increase in hair length produces a larger increase in uptake than an 8-fold increase in hair density.

Fig. 3 gives the effects of the same combinations of L_h and N on uptake after 14 d calculated with the three models at three moisture contents. The concentration–distance curves for different moisture contents have the same form as those in Fig. 1, although with different scales on the y -axis, so are not shown. Fig. 3 shows that there are differences among the three models. The relative increase in uptake with hair density (seen in Fig. 2) is greatest with Single 2 and comparable for Single 1 and the dual porosity model. Also, the relative decrease in uptake with decrease in moisture content is least with Single 2 and most with Single 1.

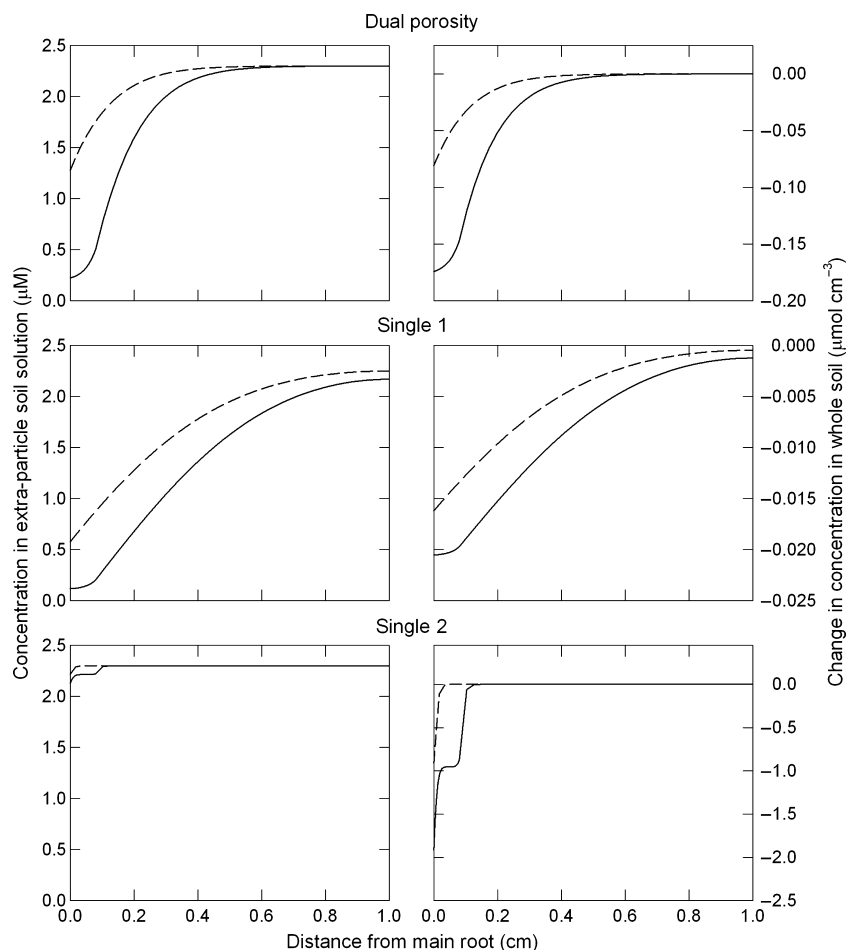


Fig. 1 Phosphate concentration profiles with distance from the main root axis, with (solid line) and without (dashed line) root hairs, calculated with the dual and single porosity models. Panels on the left show concentrations in the extra-particle solution; panels on the right are changes in the whole soil concentration. In Single 1, the concentration on external soil surfaces is as in the dual porosity model; in Single 2, the total concentration in the soil is the same. In all cases the initial concentration in solution is the same ($= 2.3 \mu\text{M}$). Other parameter values are as given in Table 2, with water content as in Case 1 in Table 3 ($\theta = 0.471$; $f_e = 0.366$) and $t = 14$ d.

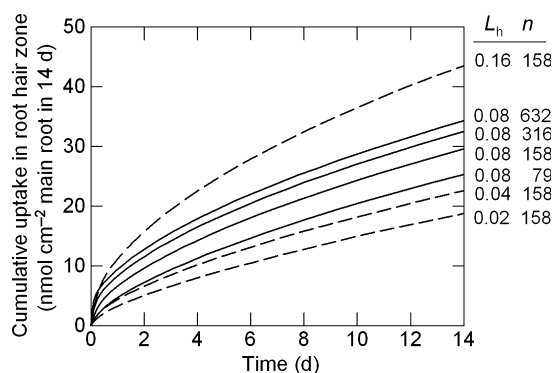


Fig. 2 Effects of root hair length (L_h) and density (N) on phosphate uptake calculated with the dual porosity model. $L_h = 0.08$ cm at $N = 79, 158, 316$ and 632 cm^{-1} (solid lines) and $L_h = 0.02, 0.04$ and 0.16 cm at $N = 158 \text{ cm}^{-1}$ (dashed lines). Other parameter values are as given in Table 2 with water content as in Case 1 in Table 3 ($\theta = 0.471$; $f_e = 0.366$).

These differences reflect the different amounts of plant-available phosphate with the three models and resulting differences in depletion rates around the root.

Fig. 4 gives the results of varying L_h at constant total mass of root hairs ($= N \times \rho \times \pi a_h^2 L_h$, where ρ is the density

of root hair tissue). For a given mass of root hair, Fig. 4 shows that uptake increases with hair length with the dual porosity and Single 1 models, but it decreases with hair length with Single 2. The effect of moisture content is marked in Single 1, and to a lesser extent with the dual porosity model, but only slight with Single 2. Also with Single 1 and, especially, the dual porosity model, uptake is not very sensitive to root hair density and there appears to be a critical density above which there is little benefit to the plant in further increases. The effects depend on moisture content. With the dual porosity model, an 8-fold increase in L_h at $N = 158 \text{ cm}^{-1}$ produces a 2.0-fold increase in uptake at $\theta = 0.47$ but a 3.5-fold increase at $\theta = 0.20$, whereas an 8-fold increase in density N at length $L_h = 0.02$ or 0.16 cm produces a 1.3-fold increase in uptake at $\theta = 0.47$ but only a 1.2-fold increase at $\theta = 0.20$.

Discussion

The simulations presented in Figs 1–4 show that the representation of soil micro-structure in models significantly affects predictions of nutrient uptake by roots and root hairs. For the single porosity model with the same

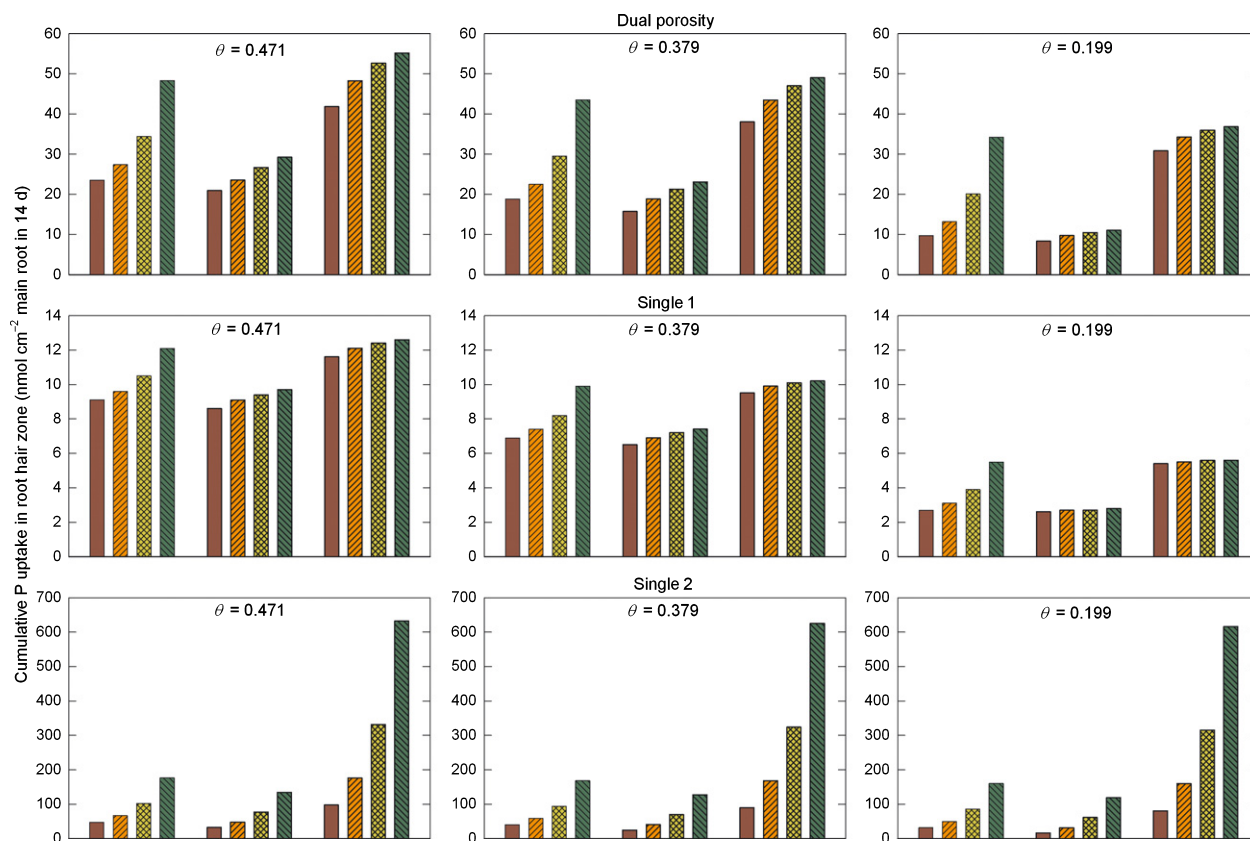


Fig. 3 Effects of root hair length (L_h) and density (N), and soil moisture content (θ) on phosphate uptake calculated with the dual porosity model compared with the two single porosity models. In each panel there are three sets of bars: the left-hand set shows $L_h = 0.02, 0.04, 0.08$ and 0.16 cm with $N = 158$ cm $^{-1}$; the middle set shows $L_h = 0.02$ cm with $N = 79, 158, 316$ and 632 cm $^{-1}$; the right-hand set shows $L_h = 0.16$ cm with $N = 79, 158, 316$ and 632 cm $^{-1}$. Other parameter values are as given in Table 2 and $t = 14$ d. Note the different y-axis scales for the three models.

phosphate concentration on external particle surfaces as in the dual porosity model, but no access to phosphate on internal surfaces, exhaustion of phosphate within the root hair zone is rapid and uptake is then limited by diffusion from the soil bulk into the root hair zone, as shown by the concentration–distance profiles in Fig. 1. For a given mass of root hair, uptake therefore increases with root length because the radius of the effective root hair zone increases, and the effect is increasingly important at low moisture content because diffusion is increasingly limiting. In contrast, with the single porosity model with the same total phosphate concentration as in the dual porosity model, but all of it on external particle surfaces, there is ample phosphate for uptake within the root hair zone. Uptake is therefore limited by the root hair surface within the root hair zone, and, for a given mass of root hair, it decreases with root length because the root hair density decreases, and it is relatively insensitive to soil moisture. The results for the dual porosity model are intermediate between the two single porosity models. Uptake is three or four times greater than with Single 1, as a result of the phosphate supply from within soil particles, and it is less dependent on soil mois-

ture content, because, as simulated, this affects diffusion through the macro-pores between soil particles but not diffusion within the particles.

The increased effect of root hairs at lower moisture content is consistent with the known proliferation of root hairs under low soil moisture conditions observed in a range of crop plants (Mackay & Barber, 1985; Sangakkara *et al.*, 1996). Also, Pariasca-Tanaka *et al.* (2009) found that phosphate-deficiency tolerant rice (*Oryza sativa*) lines in a recombinant inbred population had longer root hairs than the susceptible lines and the differences were more marked at lower moisture contents.

The results for hair length and density indicate that length is more important than density under phosphate-limited conditions, and there appears to be a critical density above which there is little benefit to the plant in further increases. The reason for this can be seen in Eqn 20: if the density, N , becomes very large, the sink term for the effect of hairs dominates Eqn 20, and the phosphate concentration in the soil solution in the hair zone then decreases towards the minimum value specified by the Michaelis–Menten uptake kinetics, at which point further increases in N will have no

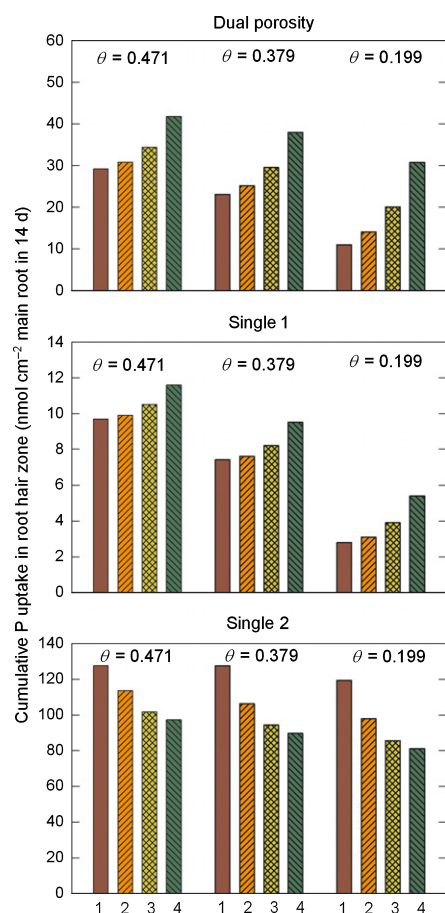


Fig. 4 Effects of root hair length (L_h) and density (N), and soil moisture content (θ) on phosphate uptake calculated with the dual porosity model compared with the two single porosity models. In each case the mass of root hairs per unit main root is the same, but L_h (cm) and N (cm^{-1}) vary: (1) 0.02 and 632, (2) 0.04 and 316, (3) 0.08 and 158, and (4) 0.16 and 79, respectively. Other parameter values are as given in Table 2 and $t = 14$ d.

effect. The greater effect of hair length than density agrees with the experimental findings of Bates & Lynch (2000), who compared phosphate uptake by *Arabidopsis* mutants with defects in root hair density and elongation. Under phosphate-limiting conditions, uptake per unit root length by the mutant with impaired hair density, but the same elongation, was 63% that of the wild type. However, for the mutant with impaired hair elongation, but the same density, uptake was only 33% of that by the wild type. This order of effect is consistent with the predictions of the dual porosity model. The single porosity models failed to predict these effects.

The model also shows that the effects of length and density on nutrient capture depend on the surrounding soil moisture content. The relative increase in uptake with density is smaller in dry soil, presumably because the minimum concentration for uptake in the soil solution in the root hair zone is reached more rapidly. By contrast, the rela-

tive increase with hair length is greater in dry soil. We have not attempted to explicitly link root hair geometry to changes in soil moisture or phosphate concentration. However, to do so would be straightforward, given the necessary experimental data from which to derive relationships. Root hair density N is both physically and genetically constrained (see the Introduction). Cellular patterning within the root apex and cellular expansion in the elongation zone (approx. 0.2–1.0 cm back from the apex) will ultimately determine the maximum value of N (Savage & Schmidt, 2008). It should be noted, however, that, under some conditions, post-embryonic reprogramming of rhizodermal cells to root hairs can occur (e.g. under manganese deficiency) which could lead to a change in N and a mixed population of root hair lengths (Yang *et al.*, 2008). Current evidence also suggests that N is regulated at the whole-plant level, leading to a uniform spatial pattern along the root while hair length can be regulated by more localized soil conditions leading to a relatively constant hair density but a mixed population of hair lengths (Gilroy & Jones, 2000; Savage & Schmidt, 2008). The dual porosity model provides a means of assessing the consequences of these adaptations for nutrient capture.

Although not considered explicitly in the model here, it is possible that root hairs actively manipulate soil moisture contents in the rhizosphere to promote nutrient capture, either by secretion of exopolysaccharides (EPS) or by efflux of water from the root. Secretion of mucilage or water facilitates the formation of water bridges with soil particles and thus increases the effective diffusion coefficient of nutrients moving towards roots. This response is particularly pronounced in dry soil (Watt *et al.*, 1994). Current evidence suggests that, while relic mucilage may be present at the base of root hairs, root hairs do not themselves release significant amounts (Willats *et al.*, 2001). However, they may stimulate EPS-producing bacteria, which are abundant in most soils (Trollenier & Hecht-Buchholz, 1984; Peterson & Farquhar, 1996). Assuming the effect of mucilage secretion in the root hair zone is effectively to increase the soil extra-particle solution volume, and that the diffusion coefficient D_1 is unchanged, then the dual porosity model predicts that mucilage secretion sufficient to double the extra-particle solution volume at the low moisture contents in our simulations would increase phosphate uptake 2.5-fold. Further model development is needed to explore these effects.

The experimental evidence on interactions between root hair length and density, soil conditions and nutrient uptake is thin, as discussed in the Introduction section. However, new imaging and other techniques for measuring roots and rhizosphere processes are making experiments on these topics increasingly possible (Darrah *et al.*, 2006; Gregory *et al.*, 2009; Dupuy *et al.*, 2010). The model presented here, which is based on first principles with no arbitrary

assumptions about underlying mechanisms, will allow such measurements to be properly exploited.

Conclusions

- The dual and single porosity models gave significantly different results.
- For a given nutrient concentration in the soil solution and given buffering on external particle surfaces, the dual porosity model predicted greater root uptake because of slow release of nutrient from within soil particles. It also predicted a smaller effect of soil moisture.
- The dual porosity model predicted greater increases in uptake with increases in root hair length than with increases in root hair number per unit main root length; the single porosity model did not. Also, the effect of hair length was less sensitive to soil moisture content. The effect of hair density was less in dry soil.
- The predictions of the dual porosity model are more consistent with experimental observations than those of the single porosity model.

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Supporting Information

Additional supporting information may be found in the online version of this article.

Fig. S1 Solution of the cell problem, calculated using COMSOL MULTIPHYSICS.

Fig. S2 Geometry of the model.

Notes S1 Derivation of the model: nondimensionalization.

Notes S2 Derivation of the model: multiscale expansions.

Notes S3 Derivation of the model: calculation of the effective diffusivity.

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